How much can AI add to productivity and GDP in Latin America and the Caribbean?

Methodological note

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Abstract

This note documents an upper bound on the ten-year effect of artificial intelligence on total factor productivity (TFP) and GDP for 22 countries of Latin America and the Caribbean, built only from public data. It applies the task-based accounting of Acemoglu (2025), calibrated as in Aghion and Bunel (2024), country by country, and adds a structural-transformation block in the spirit of Baumol (1967), following Huneeus and Rogerson (2024) and Comin, Lashkari and Mestieri (2021), to ask how the bound changes when sectoral weights move with observed productivity trends. The benchmark GDP gain after ten years ranges from 1.65% to 3.27%, 44% to 87% of the bound for the United States under the same calibration. With Acemoglu's own calibration the bound is 20% to 29% of that. The note states every assumption, separates what is taken from the literature from what is constructed here, and lists the limitations.

1 Overview

The exercise has two parts. Part (a) is an accounting bound: the aggregate gain from AI is the GDP weight of the tasks AI makes cheaper times the cost saving on them (Hulten's theorem). Part (b) recognizes that the weights of that sum are sectoral shares, and that those shares respond to sectoral productivity: with complementary sectors, the sectors whose productivity grows fastest lose nominal weight. Part (b) recomputes the bound with the sectoral weights that observed productivity trends produce ten, twenty and thirty years ahead.

The numbers are level effects cumulated over ten years, not growth rates, and they are upper bounds in the sense of Acemoglu (2025): they assume that every exposed task that is profitable to automate is in fact automated within ten years. They are about existing tasks. New tasks, new products and effects on the production of ideas are outside the bound; Section 6.1 explores one such channel as an extension.

2 The accounting

Following equation (14) of Acemoglu (2025), the TFP gain over ten years in country c is

$$\Delta \ln \mathrm{TFP}_c = e_c \times q_c \times a \times s_{L,c}, \tag{1}$$

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where e_c is exposure (the share of the wage bill in tasks AI could perform), q_c is profitability (the share of exposed tasks that are worth automating within ten years), a is the labor-cost saving per automated task and $s_{L,c}$ is the labor income share, which converts a wage-bill share into a GDP share. Capital is supplied elastically at a fixed real rental rate, so the capital-output ratio is constant and

$$\Delta \ln \mathrm{GDP}_c = \frac{\Delta \ln \mathrm{TFP}_c}{s_{L,c}} = e_c \times q_c \times a. \tag{2}$$

The labor share therefore scales TFP and cancels in GDP. Acemoglu's equation (15) separates hard-to-learn tasks, with a lower saving; it is used as a robustness check (Section 6).

3 Data

All inputs are public and comparable across countries.

- **ILOSTAT** (SDMX API): employment by occupation at one and two digits of ISCO-08; employment by economic activity (ISIC Rev. 4 sections) and by activity and occupation; mean monthly earnings of employees by occupation and skill level in US dollars and local currency, for each country and for the United States; labor income share in GDP (SDG 10.4.1, with imputation of the labor income of the self-employed).
- **CEPALSTAT** (indicators 2195, 2196 and 2216): value added by activity at current and constant prices.
- **Task exposure**: the scores of Eloundou et al. (2024) for 19,265 O*NET tasks, as used by Acemoglu (2025), mapped to ISCO-08.
- **Adoption cost**: the cost model of Svanberg et al. (2024) for computer-vision systems.
- **OECD TiVA 2025**: gross output and value added by industry, for the input-output decomposition.
- **Employer social security rates**: Ius Laboris (2017) for South America; US Social Security Administration, *Social Security Programs Throughout the World: The Americas, 2019* for Mexico, Central America and the Caribbean.

Chile is also computed with household-survey microdata (CASEN 2022) and national accounts of the Central Bank; that detailed calculation is used here only as a check on the cross-country method.

4 Part (a), country by country

4.1 Exposure e_c

Each occupation's exposure is the US task-level exposure of Eloundou et al. (2024), translated from SOC to ISCO-08 at four digits and averaged to two digits (one digit where the country only publishes major groups) with Chilean wage-bill weights from CASEN. Countries differ only in the weights: the country's employment by occupation (latest year up to 2024) times the relative earnings of its major group. Exposure ranges from 14% to 20% of the wage bill, against 20.0% in the United States. As a check, Chile is 16.7% with this method and 17.4% with CASEN microdata. The assumption that the task content of an occupation is the same as in the United States is the main limitation of this step.

4.2 Profitability q_c

Svanberg et al. (2024) estimate that at 2023 costs it is profitable to automate 23% of exposed (computer-vision) tasks in the United States, and report how that share changes with the cost of the AI system relative to the labor it replaces (their Figure 6b). Let m_c be the ratio of labor cost to system cost in country c relative to the United States:

$$m_c = \frac{\omega_c \, r_c / r_{US}}{\lambda \, \omega_c^P \, r_c / r_{US} + (1 - \lambda)}, \tag{3}$$

where ω_c is the ratio of mean monthly earnings in dollars of employees in exposed occupations to the US value (by ILO skill level, weighted by the country's exposed employment), r_c is the employer surcharge (Chile's measured value scaled by each country's statutory employer contribution rate), ω_c^P is the same earnings ratio for professionals (skill levels 3–4) and $\lambda = 42\%$ is the share of the system cost that is in-house engineering in Svanberg et al.'s cost structure. The remaining $1 - \lambda = 58\%$ (data, compute, external vendor) is paid at US prices in every country. Labor in exposed occupations costs 5% to 45% of the US level, while the AI system costs 60% to 74%.

Acemoglu calibration. Reading the Svanberg curve $F(\cdot)$ at m_c, normalized to 23% for the United States, gives $q_c^A = 0.23 \, F(m_c)/F(1)$: 9% to 19%.

Benchmark (Aghion–Bunel calibration). Acemoglu holds the 23% constant for ten years. Aghion and Bunel (2024) point out that Svanberg et al.'s own scenarios let system costs fall, and that with a 22% annual decline (their reading of the trend in GPU price-performance in Hobbhahn and Besiroglu, 2022) the US share reaches 50% in ten years. The benchmark applies that logic to every country: the system cost falls by the factor $D = 0.78^{-10} \approx 12$, each country is read on the same curve at $m_c D$, and the level is anchored at Aghion and Bunel's 50% for the United States:

$$q_c = 0.50 \times \frac{F(m_c D)}{F(D)}. \tag{4}$$

This gives 25% to 45%. Two remarks. First, the country-specific adjustment is a construction of this note, not of Aghion and Bunel, who work with the United States only. Second, the curve is tabulated up to a cost ratio of 8 and is extrapolated linearly in $\log m$ beyond it; read directly it gives 46% for the United States at D, and it is anchored at 50%. Relative wages are held at current levels.

4.3 Cost saving a

Acemoglu uses $a = 27\%$, the mean of two experiments (Noy and Zhang, 2023; Brynjolfsson, Li and Raymond, 2025). Aghion and Bunel add the coding experiment of Peng et al. (2023) and use 40%, which is the benchmark here, the same in every country.

4.4 Exposure is not taken from Aghion and Bunel

Aghion and Bunel use an exposure of 60% for the United States, from the occupation-level index of Pizzinelli et al. (2023). That index counts occupations touched by AI, including complementarity; it is not a share of tasks AI can perform. The task-based measure is kept. As a consequence the benchmark for the United States under the same calibration is a TFP gain of 2.1% and a GDP gain of 3.7% over ten years, about a third of Aghion and Bunel's own 6.8% of TFP.

4.5 Labor share

$s_{L,c}$ is the latest observed ILOSTAT labor income share (0.34 to 0.58). It affects TFP and not GDP.

5 Part (b): structural transformation

5.1 Model

There are eleven sectors: agriculture and ten non-agricultural sectors built from ISIC sections. Preferences are those of Huneeus and Rogerson (2024), with a subsistence term in agriculture, and the non-agricultural aggregate is the non-homothetic CES of Comin, Lashkari and Mestieri (2021):

$$U = \alpha_a \ln(c_a - \bar{c}_a) + (1 - \alpha_a) \ln C, \qquad s_i = \Omega_i \left(\frac{p_i}{E_n} \right)^{1-\sigma} C^{\varepsilon_i}, \quad \sum_{i \neq a} s_i = 1, \tag{5}$$

where s_i is the share of sector i in non-agricultural spending E_n. With $\sigma < 1$ sectors are complements: a sector whose productivity grows faster becomes cheaper and loses nominal weight (the Baumol, or substitution, effect). The income effects come from $\bar{c}_a$ and from the elasticities ε_i. Each sector produces with a Cobb–Douglas technology in capital and labor with labor share θ_i; labor is mobile across sectors; capital is rented at a fixed real price. Sector prices are $p_i = w^{\theta_i}/A_i$, and the numéraire is the chained (Törnqvist) price index. To first order the model reproduces the accounting of Part (a).

5.2 Calibration

$\sigma = 0.35$ and $\alpha_a = 0.02$ are from Huneeus and Rogerson (2024). The income elasticities ε_i are those of Comin, Lashkari and Mestieri (2021, world sample, manufacturing normalized to one), which range from 0.08 in mining to 1.73 in finance and business services. Ω_i, the relative wages by sector and $\bar{c}_a$ reproduce each country's sectoral shares of nominal value added and employment in the base year (2022). Sectoral labor shares are Chile's, rescaled to each country's aggregate labor share. No parameter is fitted to the paths described below.

5.3 Simulation and outputs

Each sector's productivity grows at its observed trend: the annual growth of real value added per worker over the longest window available within 2013–2025 (at least six years). The AI gain of Part (a) enters sector by sector, as the sector's exposure times $q_c \times a$ times its labor share, scaled so that with 2022 weights it adds up to the aggregate bound, and is phased in linearly over ten years. The model is solved year by year for thirty years, with and without AI. The outputs are: (i) the bound recomputed with the sectoral weights of 2032, 2042 and 2052; (ii) the exact GDP effect in the model; (iii) the reallocation of workers across sectors. The model is solved at the benchmark shock and, separately, at Acemoglu's.

The block is run for 13 countries. It is not run where the window is too short, where employment by sector is missing, or where the model does not converge on the observed trends (Guyana's oil boom, Barbados). Figure-level evidence on the Baumol mechanism uses the 18 countries with at least four years of sectoral data.

5.4 Validation

For each country the model is run over its own historical window with observed productivity growth, and predicted changes in employment shares are compared with actual ones. Correlations range from 0.37 to 0.94.

5.5 What the block shows

The change in the bound from using future weights is a covariance between the drift in weights and the sectoral gains. It is -0.49 to $+0.07$ percentage points of GDP over twenty years: a few percent of the gain in most countries and 23% in Mexico, where the exposed services have been the fast-productivity sectors. The relative erosion does not depend on the size of the AI shock, because the weights follow historical trends, not AI. AI itself reallocates at most 0.25 of every 100 workers across sectors in a decade, against 2.6 to 9.4 from the trends.

6 Robustness and extensions

- **Acemoglu calibration** (q_c^A, $a = 27\%$): GDP gain 0.33% to 0.93%, 20% to 29% of the benchmark.
- **No cost gap with the United States** ($q = 50\%$ everywhere): 2.8% to 4.0%.
- **Hard-to-learn tasks** (Acemoglu's equation 15, with the saving on hard tasks in the same proportion to the saving on easy tasks as in his paper): about 21% lower.
- **CES between capital and labor** ($\sigma_{KL} = 0.7$, Oberfield and Raval, 2021): the local multiplier from TFP to GDP becomes $1 + \sigma_{KL} s_K / s_L$ instead of $1/s_L$; model GDP is 12% to 24% below the bound, against 1% to 11% with Cobb–Douglas. The TFP gain and the Baumol drift are unchanged.
- **Input-output links** (Filippucci, Gal and Schief, 2024, with OECD TiVA): the bound decomposes into a direct effect (56% to 68%) and a pass-through of cheaper inputs. The two add up to the bound, so this is a decomposition and not an additional gain. Available for 5 countries.
- **Institutional adoption costs**: cybersecurity, data governance and compliance are not in the Svanberg cost model. As an illustration, not an estimate, doubling the system cost lowers the bound by 15% to 24%.

6.1 Scale effects through ideas

The bound is about existing tasks. One channel it leaves out is the production of ideas. In a semi-endogenous growth model (Jones, 1995; Aghion, Jones and Jones, 2019), the stock of ideas A, measured in TFP units, evolves as

$$\frac{\dot{A}}{A} = \theta \, S^{\lambda} A^{-\beta}, \tag{6}$$

where S is effective research input. Bloom, Jones, Van Reenen and Webb (2020, Table 7) estimate $\beta = 3.1$ for the aggregate US economy with $\lambda = 1$ (and 2.4 with $\lambda = 0.75$). If AI lowers the cost of exposed research tasks as it does for other tasks, a dollar of research buys $\Delta = -\ln(1 - e\,q\,a) = 4.1\%$ more effective input at the frontier ($e = 20\%$, $q = 50\%$, $a = 40\%$). The long-run effect is a level increase of TFP of $\lambda\Delta/\beta = 1.3\%$, not a permanent increase in

growth. Log-linearized, the gap closes at rate βg per year, with $g = 1\%$ the steady-state growth of TFP as in Jones (2022). With adoption phased in linearly over ten years, TFP is 0.18% higher after ten years and 0.49% after twenty. Adding this to the bound raises the ten-year GDP gain by 0.3 to 0.5 percentage points (11% to 25% of the benchmark).

This is an exploration. It assumes that frontier ideas reach every country one for one and without delay, which is an upper bound for the region; that research tasks are no more exposed than the average task; and it excludes product variety, new tasks, and the automation of idea production itself, which could raise growth permanently.

7 Results

Table 1 reports the inputs and results by country.

Table 1: Inputs and results by country. Gains are cumulative over ten years, in percent.

			q (%)			Benchmark		GDP, variants		Structural transformation			
Country	e (%)	m	Acem.	Bench.	s_L	TFP	GDP	Acem.	Scale	Window	Valid.	Eros. (%)	pp
Bahamas	18.1	0.61	19.0	45.0	0.34	1.12	3.27	0.93	3.80	–	–	–	–
Barbados	18.8	0.44	16.6	41.9	0.55	1.74	3.15	0.84	3.48	–	–	–	–
Costa Rica	18.5	0.31	14.1	38.3	0.55	1.57	2.83	0.70	3.17	–	–	–	–
Argentina	20.2	0.20	11.4	34.3	0.50	1.39	2.77	0.62	3.14	2013–2025	0.37	−0.3	−0.01
Uruguay	17.8	0.32	14.2	38.5	0.56	1.53	2.74	0.68	3.07	2016–2024	0.56	−0.5	−0.01
Belize	18.0	0.24	12.3	35.8	0.46	1.18	2.57	0.60	2.97	–	–	–	–
Chile	16.7	0.23	12.1	35.5	0.54	1.29	2.38	0.55	2.72	2013–2025	0.66	−6.1	−0.15
Brazil	17.4	0.18	10.7	33.1	0.58	1.33	2.30	0.50	2.62	2013–2025	0.69	+3.1	+0.07
Jamaica	18.3	0.14	9.1	30.2	0.54	1.19	2.21	0.45	2.55	–	–	–	–
Mexico	16.8	0.16	10.1	32.2	0.36	0.79	2.16	0.46	2.67	2013–2025	0.90	−22.5	−0.49
Panama	15.0	0.23	12.1	35.6	0.35	0.75	2.14	0.49	2.66	2018–2025	0.94	−8.5	−0.18
Colombia	17.3	0.14	9.4	30.8	0.51	1.08	2.13	0.44	2.49	–	–	–	–
El Salvador	16.8	0.15	9.7	31.3	0.51	1.07	2.11	0.44	2.47	2013–2025	0.66	−6.1	−0.13
Dominican Republic	17.2	0.13	9.0	30.2	0.48	0.99	2.08	0.42	2.46	2018–2025	0.89	+0.4	+0.01
Guatemala	16.6	0.15	9.5	31.0	0.44	0.90	2.06	0.42	2.48	2013–2025	0.87	−6.8	−0.14
Peru	16.2	0.14	9.2	30.6	0.43	0.85	1.98	0.40	2.40	2013–2024	0.78	+3.7	+0.07
Guyana	14.9	0.18	10.5	32.9	0.55	1.08	1.96	0.42	2.30	–	–	–	–
Honduras	16.2	0.13	9.0	30.1	0.57	1.12	1.95	0.39	2.27	2015–2025	0.88	−4.0	−0.08
Suriname	19.1	0.09	8.6	25.4	0.50	0.97	1.94	0.44	2.31	–	–	–	–
Paraguay	15.8	0.12	8.6	29.1	0.45	0.83	1.84	0.37	2.24	–	–	–	–
Ecuador	14.4	0.15	9.6	31.2	0.57	1.03	1.80	0.37	2.12	2013–2025	0.48	−10.7	−0.19
Bolivia	13.8	0.13	8.9	29.9	0.53	0.87	1.65	0.33	2.00	2017–2025	0.72	−7.5	−0.12
United States	20.0	1.00	23.0	50.0	0.57	2.13	3.74	1.16	4.06	–	–	–	–

Notes: e exposure; m labor cost relative to AI-system cost (US = 1); q profitability under Acemoglu's calibration (2023 costs) and the benchmark (costs falling 22% a year for ten years); s_L labor income share. "Acem." is the GDP gain with Acemoglu's calibration; "Scale" adds the ideas channel of Section 6.1. Window: years used for sectoral productivity trends. Valid.: correlation between predicted and actual changes in employment shares over the window. Eros.: change in the bound with 2042 instead of 2022 sectoral weights, in percent of the gain and in percentage points of GDP. "–": not computed. The US row uses Acemoglu's exposure; its Acemoglu-calibration entry is his published number.

8 Limitations

1. The bound assumes full adoption of profitable tasks within ten years and ignores adjustment, organizational and institutional costs beyond the Svanberg cost model.
2. Task exposure by occupation is that of the United States; only the occupational mix varies by country, at one or two digits of ISCO-08.
3. The Svanberg cost curve is for computer vision; it is applied to all exposed tasks, as in Acemoglu and in Aghion and Bunel. Its extension to country-specific cost ratios after ten

years of falling costs requires extrapolating the curve.

4. Earnings are monthly, not hourly; employer surcharges are statutory rates, and Mexico's is an estimate at the average wage because its schedule is not a flat rate.
5. Sectoral labor shares are Chile's, rescaled; the labor share of ILOSTAT imputes self-employment income with methods that differ across countries.
6. The structural block projects observed trends of at most twelve years forward for up to thirty; windows that include the pandemic are noisy, and the result for Mexico is large relative to the other countries and should be read with that in mind.
7. Countries are left out where data are missing: Venezuela, Nicaragua, Trinidad and Tobago, Haiti and Cuba from the accounting; a further 9 countries from the structural block.
8. The scale-effects extension is illustrative and rests on full and immediate diffusion of frontier ideas.

9 Replication

The calculation is a chain of Python scripts. Data download: `s1_datos_sudamerica.py`, `s2_datos_latam.py`. Accounting and structural block by country: `e17_latam.py`, with the model in `modelo_ce.py`. Input-output decomposition: `e18_insumo_producto_latam.py`. The presentation, this note and the interactive calculator are generated from the same output files by `p3_academica_en.py`, `nota_metodologica.py` and `build_dashboard.py`. Code and data are available on request.

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